

AMENDMENT NO. 1 JULY 2016
TO
IS 12222 : 2011 AUTOMOTIVE VEHICLES — TURNING CIRCLE AND
MANEUVERABILITY REQUIREMENTS

(First Revision)

(Second cover page, Foreword, para 2) — Insert the following after the second para:

This standard has been now amended to include the maneuverability requirements for single-deck rigid or articulated vehicles designed and constructed with more than two axles for carrying in excess capacity with more than 22 passengers, whether seated or standing in addition to the driver and having an overall width exceeding 2.3 m. This has been done based on the following assessment carried out by the working group/panel of this committee:

- a) The differences in allowable axle weights in EU nations vis-à-vis legally permissible safe axle weights in India and its inter-relationship on the turning circle limit.
- b) Review of applicable international standards and the deviations taken by ECE/EU countries and countries that recognize ECE approvals.
- c) Introduction of performance limit for turning circle radius in addition to the proposed limit for turning clearance circle radius for specific vehicle categories.
- d) Confirmation from Central Road Research Institute (CRRI) and the data on recommendations of Indian Road Congress (IRC) on road width and minimum turning radius in road construction.
- e) Safety analysis data subsequent to the route survey carried out for a typical large passenger vehicle on a specifically identified route to determine the safety, risk assessment and to evaluate the steerability.
- f) Need for mass transportation of passengers with optimised per capita fuel consumption.

Based on the above factors the panel decided to consider the outer circle requirements of 14 m radius against 12.5 m radius in ECE, which is a deviation from ECE.

Based on the above approach the panel decided to consider the outer circle requirement of 14 m radius against 12.5 m radius in ECE, for single decked or articulated passenger vehicles having more than 22 seated passengers and more than two axles, which is a deviation from ECE.'

(Page 1, clause 3.1) — Substitute the following for the existing clause:

'3.1 Turning Circle Diameter (Outer) — This shall be the outer diameter or outer turning circle defined in 5.31 of IS 9435.'

(Page 1, clause 3.2) — Substitute the following for the existing clause:

'3.2 Turning Clearance Circle Diameter (Outer) — This shall be the diameter of the circle as defined in 5.32 (b) of IS 9435.'

(Page 1, clause 5.1) — Substitute the following for the existing clause:

'5.1 The vehicle shall be able to maneuver on either lock inside a circle of radius R as shown in Fig. 1 or Fig. 2 as applicable without any of its outermost points projecting outside the circumference of the circle.'

(Page 1, clause 5.2) — Substitute the following for the existing clause:

'5.2 When the outermost points of the vehicle are moving on either wheel lock on a circle of radius R (see Fig. 1 and Fig. 2) the vehicle shall be able to move within the limits of circular track ($R-r$) wide as shown in Fig. 1 or Fig. 2 as applicable.'

(Page 1, clause 6.1) — Substitute the following for the existing clause:

'6.1 The turning circle diameter (outer) as defined in 3.1 shall not exceed 24 m.'

Price Group 2

Amendment No. 1 to IS 12222 : 2011

(Page 2, Fig. 1) — Substitute the following for the existing figure:

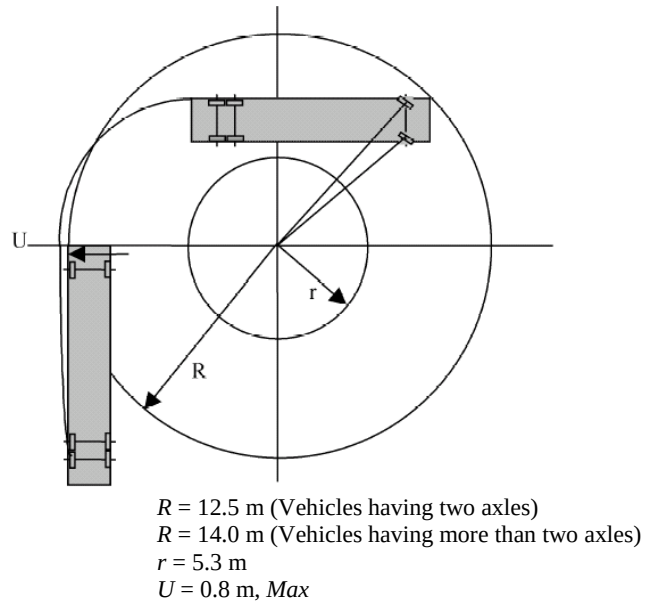


FIG. 1 MANEUVERABILITY FOR RIGID VEHICLES

(Page 2, Fig. 2) — Substitute the following for the existing figure:

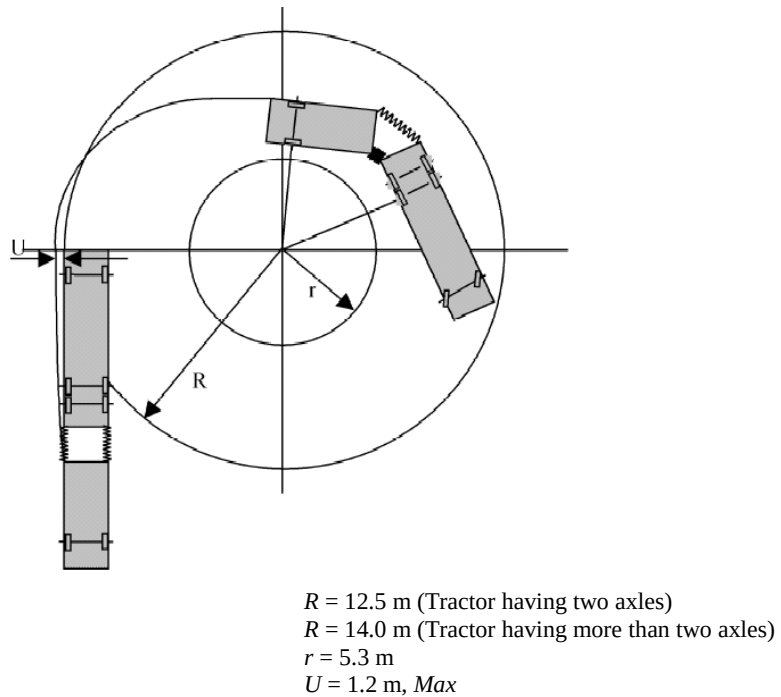


FIG. 2 MANEUVERABILITY FOR ARTICULATED VEHICLES

(Page 3, clause 6.2) — Substitute the following for the existing:

‘6.2 In the case of vehicles and tractor of articulated vehicles covered in **1.2**, the turning clearance circle diameter as defined in **3.2** shall not exceed dimension $2R$ given in Fig. 1 or Fig. 2, as applicable.’
